Product Architecture and Platform Strategy for C-SQD

1 Introduction

The central product of C-SQD is not a document reader. It is a review marketplace and scholarly evaluation infrastructure. The primary economic activity on the platform is the creation, discovery, evaluation, compensation, and aggregation of reviews. The article itself is important, but it is not the product.

This distinction has consequences for platform design. A document reader optimized for expert workflows may be desirable, but the success of C-SQD depends first on participation. Authors must be able to submit work. Reviewers must be able to accept assignments. Readers must be able to discover evaluations. Institutions must be able to assess reviewer quality and review outcomes. The platform therefore succeeds or fails based on the accessibility and visibility of its review graph rather than on the sophistication of its reading interface.

For this reason, the primary implementation should be web-native.

2 The Scholarly Object Model

The core object in C-SQD is neither a journal article nor a PDF. It is a scholarly object.

A scholarly object may correspond to a journal article, preprint, report, dataset, software package, protocol, or other intellectual contribution. Each scholarly object possesses a canonical identity through a DOI, URL, repository identifier, or similar mechanism.

Reviews attach to the scholarly object rather than to a particular hosting location. This separation is important because the platform should remain agnostic to publishers.

An article published by a major journal, a preprint hosted on a public server, and a government report should all be capable of receiving ElementReviews and SynthesisReviews. The review infrastructure therefore becomes independent of the publication venue.

This architecture also supports future migration of content. If a preprint later becomes a journal publication, the review history remains attached to the underlying scholarly object rather than being fragmented across multiple locations.

3 Article Consumption

The platform should distinguish between content that may be natively displayed and content that remains under external publisher control.

For open content, including preprints and permissively licensed publications, C-SQD may maintain a native representation of the article. This allows reviews, annotations, citations, and future analytical tooling to operate directly within the platform.

For copyrighted or publisher-controlled content, the platform should preserve the publisher as the authoritative source of the article. The reader accesses the content through the publisher while performing review activities within C-SQD.

The important observation is that both experiences lead to the same destination. Regardless of where the article is viewed, the resulting reviews are attached to the same scholarly object and become part of the same review graph.

Consequently, article presentation is a secondary implementation detail. The review infrastructure remains unchanged.

4 Why the Web Platform Comes First

A web platform minimizes participation costs.

A reviewer receiving an invitation can immediately access the assignment. An author can view evaluations without installing software. A funding agency can inspect review histories through a browser. Readers can discover reviews through links, citations, and search engines.

These properties are particularly important because C-SQD is intended to function as public scholarly infrastructure rather than as a specialized workstation.

The network effects of the platform arise from visibility. Reviews accumulate value when they become discoverable, citable, and reusable. A web-native system maximizes this visibility.

The same argument applies to unsolicited reviews. A reader who encounters a paper should be able to create a review with minimal friction. Requiring installation of a dedicated application would reduce participation precisely where network growth is most important.

5 The Economics of Review

The economic engine of C-SQD is review creation.

Paid reviews, institutional review contracts, review bounties, reviewer reputation, and future quality metrics all depend on a liquid market connecting scholarly objects with evaluators.

This market benefits from reducing barriers to entry. Every additional step between discovering a paper and contributing a review decreases participation.

The web platform therefore serves as the economic center of the system. Review assignment, compensation, dispute resolution, reviewer profiles, institutional reporting, and payment workflows should all be accessible through the browser.

The article reader supports these activities, but does not define them.

6 Scientific Bug Bounties

The same logic extends naturally to scientific bug bounties.

A bug bounty is fundamentally a review workflow. A participant identifies a potential error, provides evidence, submits an evaluation, and receives compensation if the claim is validated.

The critical requirement is low-friction participation. Researchers, readers, and domain experts should be able to enter the process directly from the scholarly object under discussion.

Because bug bounties represent another form of evaluation market, they fit naturally within the web platform and benefit from the same network effects that support ordinary reviews.

7 The Role of a Desktop Application

A desktop application may eventually provide a superior working environment for professional reviewers.

Complex review assignments often involve reading multiple papers simultaneously, comparing figures, consulting external references, managing annotations, and composing long evaluations. A dedicated workspace can improve productivity for these activities.

The desktop application should therefore be viewed as a specialized reviewer cockpit rather than as the primary product.

Its purpose is to increase the productivity of active reviewers while remaining fully synchronized with the underlying C-SQD graph.

This distinction is important. The desktop application is an interface. The platform itself remains the web-accessible review network.

8 Relationship to Observatory

Observatory occupies a different layer of the architecture.

C-SQD generates structured evaluation data. ElementReviews, SynthesisReviews, bug bounty outcomes, reviewer histories, disagreements, and corroborating assessments become components of a growing epistemic graph.

Observatory visualizes this graph.

Its purpose is to help users navigate problems, evidence, claims, methods, reviews, and disagreements. The value of Observatory therefore depends directly on the volume and quality of review activity produced by C-SQD.

The strategic relationship is analogous to infrastructure and navigation. C-SQD creates the evaluative substrate. Observatory provides the map.

9 Conclusion

The defining asset of C-SQD is not article hosting and not document viewing. It is a marketplace and infrastructure for scholarly evaluation.

This observation suggests a clear product strategy. The web platform should be treated as the primary system because it maximizes participation, discoverability, and economic activity. Native article rendering can be used when licensing permits, while externally hosted articles remain connected through canonical scholarly objects.

A desktop application may eventually become valuable for professional reviewers, but it should be understood as an optional productivity layer built on top of the underlying review network.

The long-term asset is the review graph itself. Everything else exists to increase its growth, quality, and usefulness.